

Overview of Tests of Significance in STAT 200

- ▶ Case I: One box, testing its average (percentage of 1's) against an external standard — Chapter 26.
- ▶ Case II: Two boxes, testing for the equality of their averages (percentages of 1's) — Chapter 27.

The general methodology for Cases I and II is the same.

The basic idea is to see how many SEs an observed result is from its expected value:

$$z = \frac{\text{observed result} - \text{expected result}}{\text{SE for the observed result}}.$$

Tests based on z are called **z -tests**.

The Difference between Two Chance Quantities

Let X and Y be two random quantities. Consider their difference $X - Y$.

- ▶ The **expected value** of $X - Y$ is just

$$(\text{expected value of } X) - (\text{expected value of } Y)$$

no matter X and Y are independent or not.

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- ▶ smaller than $(\text{SE of } X) + (\text{SE of } Y)$
due to some cancellation of the chance errors
- ▶ NOT applicable if X and Y are **dependent**.

Example 1

100 draws are made at random *with* replacement from the box A below. Independently, 100 draws are made at random *with* replacement from the box B.

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---	---

(B)

3	4
---	---

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- The number of $\boxed{1}$'s will be ≈ 50 , with $SE = 5$.

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- ▶ The expected value for the difference is $50 - 50 = 0$.
- ▶ Since the draws are independent, the SE for the differences is

$$\sqrt{5^2 + 5^2} \approx 7.$$

Example 2

100 draws are made at random *with* replacement from the box

$$\boxed{\boxed{1} \boxed{2} \boxed{3} \boxed{4}}.$$

- The number of $\boxed{1}$'s will be around 25, with SE equals

$$\sqrt{100} \times \sqrt{\frac{1}{4} \times \frac{3}{4}} = 4.33.$$

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- ▶ The number of $\boxed{4}$'s will be around 25, with SE = 4.33.
- ▶ The expected value for the difference is 0, but the SE is **not**

$$\sqrt{4.33^3 + 4.33^2} = 6.12,$$

because the draws are **dependent**.

Example 2 (Cont'd)

The difference between the number of 1's and 4's in 100 draws are made at random *with* replacement from the box

1	2	3	4
---	---	---	---

. is like 100 draws from the box

1	0	0	-1
---	---	---	----

- ▶ Average of box = 0, SD of box = $1/\sqrt{2} = 0.707$.
- ▶ The expected value for the difference is $100 \times 0 = 0$.
- ▶ The SE is

$$\sqrt{100} \times 0.707 = 7.07 > 6.12$$

larger than the one given by the square-root law. The number of 1's and the number of 4's are negatively correlated: if one is large, the other is likely to be small.

Technical Note

In general, when X and Y are **positively** correlated,

$$\text{SE for } (X - Y) < \sqrt{(\text{SE of } X)^2 + (\text{SE of } Y)^2},$$

because X and Y tends to move in the **same** direction and hence there tends to be **more** cancellation of the individual chance errors.

Conversely, when X and Y are **negatively** correlated,

$$\text{SE for } (X - Y) > \sqrt{(\text{SE of } X)^2 + (\text{SE of } Y)^2},$$

because X and Y tends to move in the **opposite** direction and hence there tends to be **less** cancellation of the individual chance errors.

Two-Sample z-Test

A two-sample z-test is to test whether the averages (or percentages) of two populations are the same or not.

In terms of box models, there is a box for each of the 2 populations. A simple random sample is drawn from each of the 2 boxes.

The test hypotheses are either

- ▶ H_0 : The difference of the average of the two boxes $= c$.
- ▶ H_A : The difference of the average of the two boxes $\neq c$.

for 2 general boxes, or

- ▶ H_0 : The diff. of the %'s of 1's in the 2 boxes $= c$.
- ▶ H_A : The diff. of the %'s of 1's in the 2 boxes $\neq c$.

for two 0-1 boxes.

Two-Sample z-Test (Cont'd)

The test statistics is

$$z = \frac{\text{observed difference} - \text{expected difference}}{\text{SE for the difference}}$$

- ▶ observed difference = the difference of the two estimates
- ▶ Under H_0 , expected difference = c
- ▶ SE for the difference is

$$\sqrt{\left(\text{SE for the first estimate}\right)^2 + \left(\text{SE for the second estimate}\right)^2}$$

Example: Cocaine Usage

The National Household Survey on Drug Abuse was conducted in 1974 and again in 1985. Among persons age 18 to 25, the percentage of current users of cocaine increased from 3.1% to 7.7%.

Assuming that the results are based on two independent SRSs, each of size 700, is this difference real, or a chance variation?

1974 box
| ?? 0 ?? 1 |

One ticket for each person

Tickets marked

1 if cocaine user, 0 if not

Percentage of 1's in box
unknown (estimated as 3.1%)

1985 box
| ?? 0 ?? 1 |

One ticket for each person

Tickets marked

1 if cocaine user, 0 if not

Percentage of 1's in box
unknown (estimated as 7.7%)

Example: Cocaine Usage (Cont'd)

- ▶ The difference (1985 usage - 1974 usage) in cocaine usage is estimated as $7.7\% - 3.1\% = 4.6\%$.
- ▶ The SE for the 1974 sample percentage is estimated as

$$\frac{\sqrt{0.031 \times 0.969}}{\sqrt{700}} \times 100\% \approx 0.7\%.$$

Similarly, the SE for 1985 is $\frac{\sqrt{0.077 \times 0.923}}{\sqrt{700}} \times 100\% \approx 1.0\%$.

- ▶ Since the two samples are **independent**, the square root law can be used,

$$\begin{aligned} &\text{SE for the difference of the 2 sample percentages} \\ &= \sqrt{(0.7\%)^2 + (1.9\%)^2} = 1.2\%. \end{aligned}$$

- ▶ The 95% confidence interval for the increase in cocaine usage from 1974 to 1985 is $4.6\% \pm 2 \times 1.2\%$.

Example: Cocaine Usage (Cont'd)

Hypotheses:

- ▶ H_0 : The percentages of 1's in the 2 boxes are the same.
- ▶ H_A : The percentages of 1's in the 2 boxes are different.

Test statistic: $z = \frac{\text{observed difference} - \text{expected difference}}{\text{SE for the difference}}$

- ▶ observed difference = 7.7% - 3.1%
- ▶ expected difference under $H_0 = 0\%$
- ▶ SE for the difference = 1.2% (found in the previous slide)

So

$$z = \frac{(7.7\% - 3.1\%) - 0.0\%}{1.2\%} = 3.8.$$

P-value: If H_0 is true, the chance of observing a difference more than 3.8 SEs above 0 is about $0.01\% = 1$ in 10,000!

Example 2: Comparing Two Sample Averages

Freshmen at public universities work 12.2 hours a week for pay, on average, and the SD is 10.5 hours; at private universities, the average is 9.2 hours and the SD is 9.9 hours. Assume these data are based on two independent simple random samples, each of size 1,000. Is the difference between the averages due to chance? If not, what else might explain it?

Box model:

- ▶ How many boxes? _____

In each box:

- ▶ How many tickets: One ticket per student
- ▶ Numbers on the tickets: Number of hours worked/week

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Example 2 (Cont'd)

► Hypotheses:

H_0 : Average hours worked/week is same for public/private

H_A : Average hours worked/week is different

► The samples:

Schools	Sample size	Average	SD
Public	1,000	12.2 hrs/wk	10.5 hrs/wk
Private	1,000	9.2 hrs/wk	9.9 hrs/wk

► SEs: Public = 0.332, Private=0.313, Difference=0.46 hours

► Test statistic and P -value:

$$z = \frac{12.2 - 9.2}{0.46} \approx 6.5, \quad P\text{-value} \approx 0.$$

► Conclusion: It is very unlikely that this difference is just due to chance!

Randomized Controlled Experiments

Randomized Controlled Experiments

One might attempt to analyze a randomized controlled experiment using a two-sample test as if the treatment group and the control group as two independent samples.

$$z = \frac{(\text{ave. of the trt.} - \text{ave. of the ctrl.}) - \text{exp'd diff.}}{\text{SE for the difference}}$$

However, the sample averages for the two groups are **NOT independent** since subjects in a randomized controlled experiment come from **ONE population**, but some of them are assigned to either the treatment group and others the control group.

How to estimate the SE of the difference between the averages of the two groups?

A Box Model for a Randomized Controlled Experiment

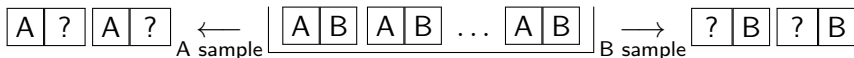
Here is a box model for a randomized controlled experiment comparing treatments A and B (e.g., a vaccine and a placebo).



There is one ticket in the box for each subject in the experiment. Each ticket has two numbers:

- ▶ one shows what the subject's response to treatment A would be,
- ▶ the other what the response to treatment B would be.

Only one of the two numbers can be observed.



- ▶ Some tickets are drawn at random *without* replacement from the box, and the responses to treatment A are observed.
- ▶ Then, a second sample is drawn at random *without* replacement from the remaining tickets, and the responses to treatment B are observed.
 - ▶ The two samples are *dependent*.
- ▶ The parameter of interest is the difference between
 - the average of the “A” numbers on all tickets in the box, i.e., what the average response to treatment A would be, if everybody were to get that treatment
 - and
 - the average of the “B” numbers on all tickets in the box.

Randomized Controlled Experiments (Cont'd)

The SE for the difference between the two sample averages can be **conservatively** estimated as follows:

- ▶ compute the SEs for each average as if the drawing are made *with* replacement;
- ▶ combine the SEs as if the two samples were **independent**.

“Conservatively” means that this procedure may overestimate the actual SE for the difference between the sample averages.

- ▶ Confidence intervals computed using the conservative SE will have more than the stated confidence level.
- ▶ *P*-values computed using the conservative SE will be larger than they actually are.

Randomized Controlled Experiments (Cont'd)

The “conservative” procedure yields nearly the right SE when:

- ▶ the sample sizes are small compared to the number of tickets, because there is little difference between drawing w/ or w/o replacement, and the dependence between the averages is small too,

or when

- ▶ the "A" and "B" numbers on the tickets are highly correlated, in particular, when the two numbers on each ticket are the same.

Randomized Controlled Experiments (Cont'd)

The normal curve can be used to figure chances about the difference between the two sample averages provided the sample sizes are sufficiently large.

Remark: The procedure applies to percentages (or rates), as well as to averages.

- ▶ Classify and count; the "A" and "B" numbers are 0's and 1's.

Example: Alexithymia

People who are unable to identify or express feelings are said to be “*alexithymic*.” Such individuals may have difficulty in relating to other people and in developing stable and satisfying relationships.

Many children who grow up in alcoholic homes are *alexithymic*.

Psychologists have noted that feelings involve both affective (subjective experience of emotion) and cognitive (intellectual understanding) components.

Example: Alexithymia (Cont'd)

Two styles of group therapy for treating alexithymic adult children of alcoholics are compared.

After agreeing to participate in the study, 50 subjects were allocated to one of two treatment groups using a randomization procedure:

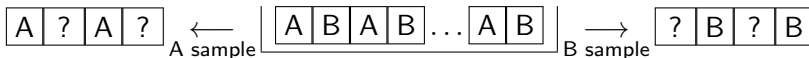
- ▶ "psychodrama" (affective emphasis) 25 subjects.
 - ▶ focus on experience.
 - ▶ reaction to events.
 - ▶ group members act out key situations in their lives.
- ▶ "choices" (cognitive emphasis) 25 subjects.
 - ▶ focus on intellectual understanding.
 - ▶ improve interpersonal communication skills.
 - ▶ lectures + group exercises.

Example: Alexithymia (Cont'd)

The extent of alexithymia was measured before and after 8 weeks of group therapy, and a measure of improvement was obtained. Values of 0 mean no improvement, and negative values indicate worsening.

	Treatment	Size	Average	SD
▶ Results:	Psychodrama	25	8	12
	Choices	25	-0.5	6

▶ Box model:



▶ 50 tickets in the box.

A = subject's score if in psychodrama group

B = subject's score if in choices group

Example: Alexithymia (Cont'd)

- ▶ The difference between what the average improvement under "psychodrama" would be for all 50 subjects and what the average improvement under "choices" would be for all 50 subjects is estimated as _____.
- ▶ To put a SE on the difference we may pretend that we have two independent samples drawn at random with replacement.
 - ▶ The SE for the "psychodrama" average is estimated as _____ = 2.4.
 - ▶ The SE for the "choices" average is estimated as _____ = 1.2.
 - ▶ The SE for the difference is (conservatively) estimated as _____ ≈ 2.7 .

Example: Alexithymia (Cont'd)

- ▶ The difference between what the average improvement under "psychodrama" would be for all 50 subjects and what the average improvement under "choices" would be for all 50 subjects is estimated as $8 - (-0.5) = 8.5$.
- ▶ To put a SE on the difference we may pretend that we have two independent samples drawn at random with replacement.
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 - ▶ The SE for the "choices" average is estimated as $6/\sqrt{25}$ = 1.2.
 - ▶ The SE for the difference is (conservatively) estimated as $\sqrt{2.4^2 + 1.2^2}$ ≈ 2.7 .

Example: Alexithymia (Cont'd)

- ▶ H_0 says no difference between the 2 therapies.
 - ▶ On this basis, the expected value for the difference between the sample averages is _____.

- ▶ The test statistic is

$$z = \frac{\text{observed diff.} - \text{expected diff.}}{\text{SE for the observed difference}} \approx \frac{8.5 - 0.0}{2.7} \approx 3.15.$$

- ▶ The normal area correspond to $z = 3.15$ is ≈ 0.9992 .
The two-sided P -value is $\approx$ _____.
The tiny P -value means the difference between the therapies is significant.
- ▶ How big is the difference?

- ▶ Is the difference big enough to be important?
- ▶ Can the results be extrapolated beyond the 50 subjects in the study?

Example: Alexithymia (Cont'd)

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- ▶ The normal area correspond to $z = 3.15$ is ≈ 0.9992 .
The two-sided P -value is $\approx 2(1 - 0.9992) = 0.0016$.
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- ▶ How big is the difference? The 95% CI for the diff. is
 $(\text{obs diff}) \pm 1.96(\text{SE of diff}) \approx 8.5 \pm 1.96 \times 2.7 \approx (3.2, 13.8)$.
- ▶ Is the difference big enough to be important?
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 $(\text{obs diff}) \pm 1.96(\text{SE of diff}) \approx 8.5 \pm 1.96 \times 2.7 \approx (3.2, 13.8)$.
- ▶ Is the difference big enough to be important?
It's a question for domain experts, not for Statisticians.
- ▶ Can the results be extrapolated beyond the 50 subjects in the study? If the 50 subjects are randomly sampled from the population of Alexithymia patients, then yes, and No if not.